

```

import java.util.ArrayList;
import java.util.InputMismatchException;
import java.util.Scanner;
import java.util.HashMap;

public class ZooApp {
    static ArrayList<Animal> animals = new ArrayList<>();

    public static void main(String[] args) {

        Scanner scanner = new Scanner(System.in);

        animals = FileManager.loadAnimals("AnimalDetails.txt"); //Makes it so it loads from
the text files on startup
        Zoo zoo = FileManager.loadZoo("ZooDetails.txt");

        int choice = 9; //Default it to 9 so even if the user puts some weird stuff later itll
restart fine.

        System.out.printf("Welcome to the system for %s in %s!\n", zoo.getName(),
zoo.getLocation());

        do {
            System.out.println("\n--- ZOO MENU ---");
            System.out.println("1. Add Animal");
            System.out.println("2. Remove Animal");
            System.out.println("3. Edit Animal");
            System.out.println("4. View Animals");
            System.out.println("5. Search Animals");
            System.out.println("6. Daily Care");
            System.out.println("7. Zoo Report");
            System.out.println("8. Update Zoo");
            System.out.println("0. Exit");

            try { //Validation for user input
                choice = Integer.parseInt(scanner.nextLine());

                if (choice >= 0 && choice <= 8) {

                    switch (choice) {

```

case 1:

addAnimalMenu(scanner);

break;

case 2:

removeAnimal(scanner, animals);

break;

case 3:

editAnimal(scanner, animals);

break;

case 4:

viewAnimals(animals);

break;

case 5 :

searchMenu(scanner);

break;

case 6 :

dailyCare();

break;

case 7:

printZooReport(animals, zoo);

break;

case 8:

editZoo(scanner, zoo);

break;

}

}

else {

```

        System.out.println("Invalid choice! Enter 0–8.");
    }
} catch (NumberFormatException e) {
    System.out.println("Please enter a valid number between 0 and 8!");
}

} while (choice != 0);

FileManager.saveZoo("ZooDetails.txt", zoo);
FileManager.saveAnimals("AnimalDetails.txt", animals); //This is what auto saves it
when the user exits.
System.out.println("File Saved!\nGoodbye!");
scanner.close();
}

public static void addAnimalMenu(Scanner scanner) {
    int type = 0;
    int age = 0;
    int fin = 0; //have to initialize them up here so the try catches work. These will not be
0 at the end anyway.

    while(true){
        System.out.println("Note: An animal with any blank data will NOT be saved when
exiting.");
        System.out.println("1. Lion 2. Eagle 3. Fish");
        try {
            type = Integer.parseInt(scanner.nextLine());

            if (type >= 1 && type <= 3) { //valid entry
                break; //move on
            } else {
                System.out.println("Enter a number between 1 and 3.");
            }

        } catch (NumberFormatException e) {
            System.out.println("Invalid input. Enter a number.");
        }
    }

    System.out.print("Name: ");
    String name = scanner.nextLine();

```

```

while (true) { //age validation
    System.out.print("Age: ");

    try {
        age = Integer.parseInt(scanner.nextLine());

        if (age >= 0) {
            break; //move on if good
        } else {
            System.out.println("Age must be 0 or higher.");
        }
    } catch (NumberFormatException e) {
        System.out.println("Invalid input. Enter a valid integer.");
    }
}

System.out.print("Color: ");
String color = scanner.nextLine();

System.out.print("Weight: ");
String weight = scanner.nextLine();

//some of the above can be left blank, but will be dealt with when writing to file
switch (type) {

    case 1 : //Lion
        System.out.print("Mane size: ");
        String mane = scanner.nextLine();
        animals.add(new Lion(name, age, color, weight, mane));
        break;

    case 2 : //Eagle
        System.out.print("Wing span: ");
        String wing = scanner.nextLine();
        animals.add(new Eagle(name, age, color, weight, wing));
        break;
}

```

```

case 3 : //Fish
    while (true) {
        System.out.print("Fin number: ");

        try {
            fin = Integer.parseInt(scanner.nextLine());

            if (fin >= 0) {
                break;
            } else {
                System.out.println("Must be 0 or higher.");
            }

        } catch (NumberFormatException e) {
            System.out.println("Invalid input. Enter a valid integer.");
        }
    }

    animals.add(new Fish(name, age, color, weight, fin));
    System.out.println("Animal added successfully.");
    break;

}
}

public static void removeAnimal(Scanner scanner, ArrayList<Animal> animals) {

    System.out.print("Enter the name of the animal you want to remove: ");
    String name = scanner.nextLine();

    boolean removed = false;

    for (int i = 0; i < animals.size(); i++) {
        Animal animal = animals.get(i);

        if (animal.getName().equalsIgnoreCase(name)) {
            animals.remove(i); //Remoevs from the array, will overwrite the file when saving.
            System.out.println("Animal " + name + " was removed successfully.");
            removed = true;
            break;
        }
    }
}

```

```

    }

    if (!removed) {
        System.out.println("No animal with the name '" + name + "' was found.");
    }
}

public static void viewAnimals(ArrayList<Animal> animals) {

    if (animals.isEmpty()) {
        System.out.println("No animals registered.");
        return;
    }

    System.out.println("\n--- ALL ANIMALS ---");

    for (Animal animal : animals) {
        System.out.println(animal.getString());
    }
}

public static void searchMenu(Scanner scanner) {
    int option = 0;
    while (true) {
        System.out.println("1. Search by name \n2. Search by color");

        try {
            option = Integer.parseInt(scanner.nextLine());

            if (option == 1 || option == 2) {
                break;
            } else {
                System.out.println("Must be 1 or 2.");
            }
        } catch (NumberFormatException e) {
            System.out.println("Invalid input. Enter a valid integer.");
        }
    }
}

```

```

if (option == 1) {

    System.out.print("Name: ");
    String name = scanner.nextLine();

    for (Animal animal : animals) {
        if (animal.getName().equalsIgnoreCase(name)) { /**This does not deal with
duplicate names, but the color one does
            System.out.println("Found: " + animal.getName());
            animal.makeSound(); // polymorphism
            return;
        }
    }

    System.out.println("Not found");
}

else if (option == 2) {

    System.out.print("Color: ");
    String color = scanner.nextLine();

    for (Animal animal : animals) {
        if (animal.getColor().equalsIgnoreCase(color)) {
            System.out.println("\n" + animal.getName());
            animal.makeSound();
        }
    }
}

public static void dailyCare() {

    for (Animal animal : animals) {

        System.out.println("\nChecking " + animal.getName());
        animal.makeSound();

        if (animal instanceof Flyable fly) { //check to see if it uses an interface, if yes do the
special method.
            fly.wingCheck();

```

```

    }

    if (animal instanceof Swimmable swim) {
        swim.waterCheck();
    }

    System.out.println("\nDaily check complete.");
}
}

public static void editAnimal(Scanner scanner, ArrayList<Animal> animals) {
    System.out.println("Note: An animal with any blank data will NOT be saved when
    exiting.");
    System.out.print("Enter name of animal to edit: ");
    String name = scanner.nextLine();

    for (Animal animal : animals) {

        if (animal.getName().equalsIgnoreCase(name)) {

            System.out.println("Editing " + animal.getName());

            //General edit for every type of animal. No special fields ye

            System.out.print("New name (leave blank to keep): ");
            String newName = scanner.nextLine();
            if (!newName.isBlank()){
                animal.setName(newName);
            }

            System.out.print("New age (-1 to keep): ");
            try {
                int newAge = Integer.parseInt(scanner.nextLine());

                if (newAge != -1 && newAge >= 0){
                    animal.setAge(newAge);
                }

            } catch (NumberFormatException e) {
                System.out.println("Invalid age, skipped.");
            }
        }
    }
}

```



```

System.out.print("New color (leave blank to keep): ");
String newColor = scanner.nextLine();
if (!newColor.isBlank()){
    animal.setColor(newColor);
}

```

```

System.out.print("New weight (leave blank to keep): ");
String newWeight = scanner.nextLine();
if (!newWeight.isBlank()){
    animal.setWeight(newWeight);
}

```

//Had to add this so that the user can also edit the special variables like mane size wingspan etc.

```

if (animal instanceof Lion lion) {
    System.out.print("New mane size(Leave blank to keep): ");
    String mane = scanner.nextLine();
    if (!mane.isBlank()){
        lion.setManeSize(mane);
    }
}

```

```

} else if (animal instanceof Eagle eagle) {
    System.out.print("New wingspan (leave blank to keep): ");
    String wing = scanner.nextLine();
    if (!wing.isBlank()){
        eagle.setWingspan(wing);
    }
}

```

```

} else if (animal instanceof Fish fish) {
    System.out.print("New fin number (-1 to keep): ");
    try {
        int fins = Integer.parseInt(scanner.nextLine());

        if (fins != -1 && fins >= 0){
            fish.setFinNumber(fins);
        }
    }
}

```

```

} catch (Exception e) {
    System.out.println("Invalid input, skipped.");
}

```

```

    }
}

    System.out.println("Animal updated.");
    return;
}
}

    System.out.println("Animal not found.");
}

public static void printZooReport(ArrayList<Animal> animals, Zoo zoo) {

    if (animals.isEmpty()) {
        System.out.println("No animals registered.");
        return;
    }

    int lions = 0;
    int eagles = 0;
    int fish = 0;

    HashMap<String, Integer> colorCount = new HashMap<>(); //dictionary to store the
    colours and their count

    for (Animal animal : animals) {

        //Get the amount of lions eagles and fish
        if (animal instanceof Lion){
            lions++;
        }
        else if (animal instanceof Eagle){
            eagles++;
        }
        else if (animal instanceof Fish){
            fish++;
        }

        //Get the different colours and their count
        String color = animal.getColor().toLowerCase();

```

```

        colorCount.put(color, colorCount.getOrDefault(color, 0) + 1);
    }

    //just get the highest colour by comparting
    String dominantColor = "";
    int max = 0;

    for (String color : colorCount.keySet()) {
        int count = colorCount.get(color);

        if (count > max) {
            max = count;
            dominantColor = color;
        }
    }

    System.out.println("\n--- ZOO REPORT ---");

    if (zoo != null) {
        System.out.println("Zoo Name: " + zoo.getName());
        System.out.println("Zoo location: " + zoo.getLocation());
    } else {
        System.out.println("Zoo Name: My Zoo");
    }

    System.out.println("Lions: " + lions);
    System.out.println("Eagles: " + eagles);
    System.out.println("Fish: " + fish);

    System.out.println("Dominant Colour: " + dominantColor);
}

public static void editZoo(Scanner scanner, Zoo zoo) {
    //this top one is for if there is no zoo name registerd. Wont really be used but oh well.
    if (zoo == null) {
        System.out.print("Enter zoo name: ");
        String name = scanner.nextLine();

        System.out.print("Enter zoo location: ");
        String location = scanner.nextLine();
    }
}

```

```
        zoo = new Zoo(name, location);
        System.out.println("New zoo created.");
        return;
    }

    //This is for editing the current one .
    System.out.println("\n--- EDIT ZOO ---");

    System.out.println("Current Name: " + zoo.getName());
    System.out.print("New name (leave blank to keep): ");
    String newName = scanner.nextLine();

    if (!newName.isBlank()) {
        zoo.setName(newName);
    }

    System.out.println("Current Location: " + zoo.getLocation());
    System.out.print("New location (leave blank to keep): ");
    String newLocation = scanner.nextLine();

    if (!newLocation.isBlank()) {
        zoo.setLocation(newLocation);
    }

    System.out.println("Zoo updated.");
}
```